There is no universal computable function

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is the author's complete diagonalisation; the statement's quantified phrasing is delivered verbatim

Editorial note. The label thm:no-univ names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Diagonalisation. For any total computable d(x) = Un'(x,x) + 1, d differs from every row Un'(k,.) at x = k.. The one-step core of the proof: totality plus computability of Un' would make d total computable and diagonally different from every row.

Theorem 0.1. *There is no universal computable function. In other words, any function* $\mathrm{Un}'(k, x)$ *which is such that if* $f(x)$ *is a total computable function, then there is a natural number* k *such that* $f(x) = \mathrm{Un}'(k, x)$ *for every* x*, is not computable.*

Proof. The proof is a simple diagonalization: if $\mathrm{Un}'(k, x)$ were total and computable, then

$$d(x) = \mathrm{Un}'(x, x) + 1$$

would also be total and computable. However, by definition, $d(k)$ is not equal to $\mathrm{Un}'(k, k)$. Hence, for every k, the values of $d(x)$ and $\mathrm{Un}'(k, x)$ differ for at least one x, namely $x = k$. □